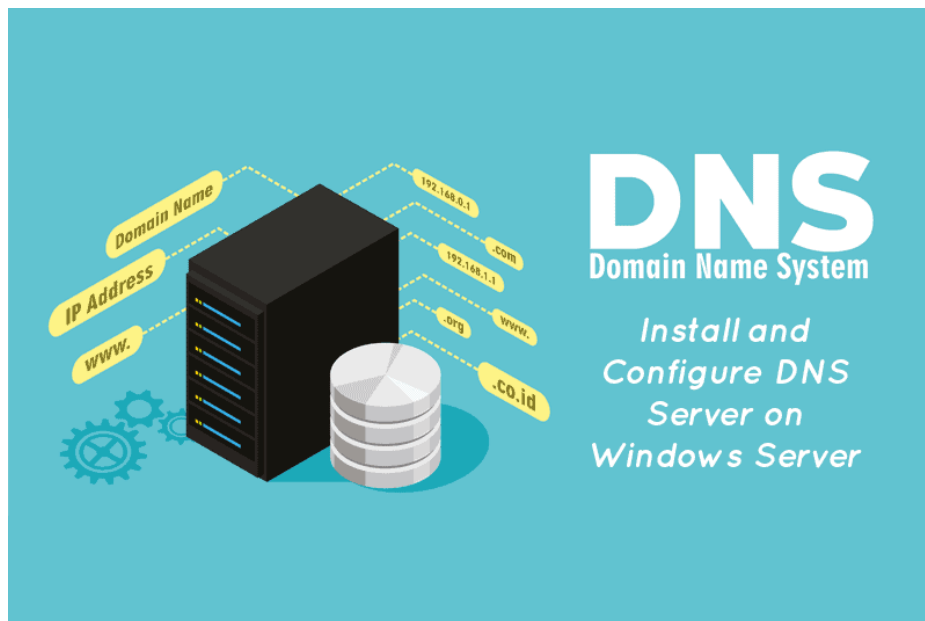


Install and configure DNS and DHCP in Server -2019-14

Introduction

DNS (Domain Name System) and DHCP (Dynamic Host Configuration Protocol) are two core network services in Windows Server.

- **DNS** translates domain names (like example.com) into IP addresses, making it easier for users to access resources without remembering numbers.



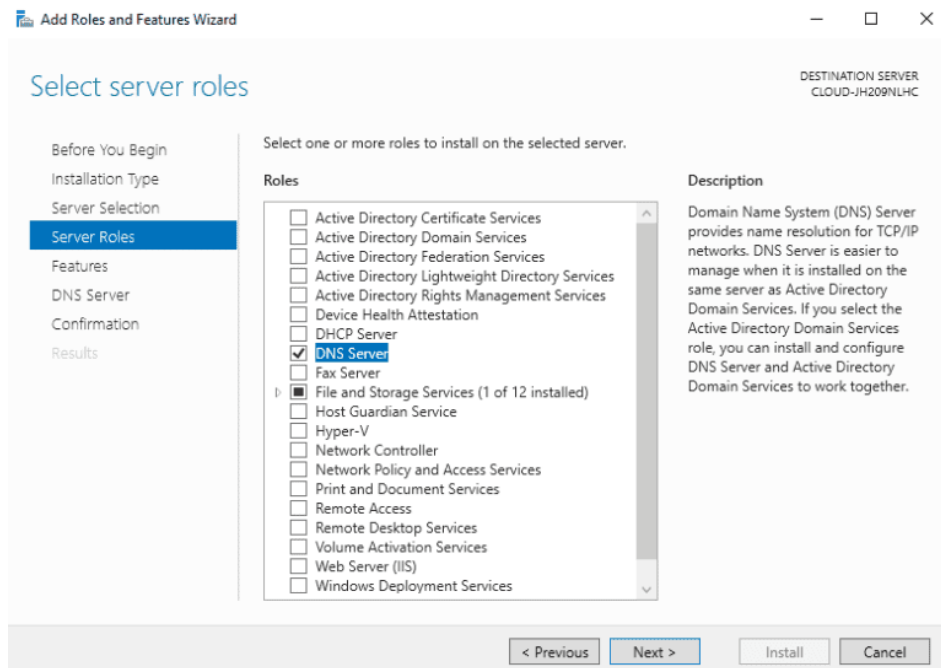
- **DHCP** automatically assigns IP addresses and other network configurations to client machines, reducing manual effort and avoiding conflicts.

Together, they provide seamless name resolution and automated IP management in a network.

Steps to Install and Configure DNS

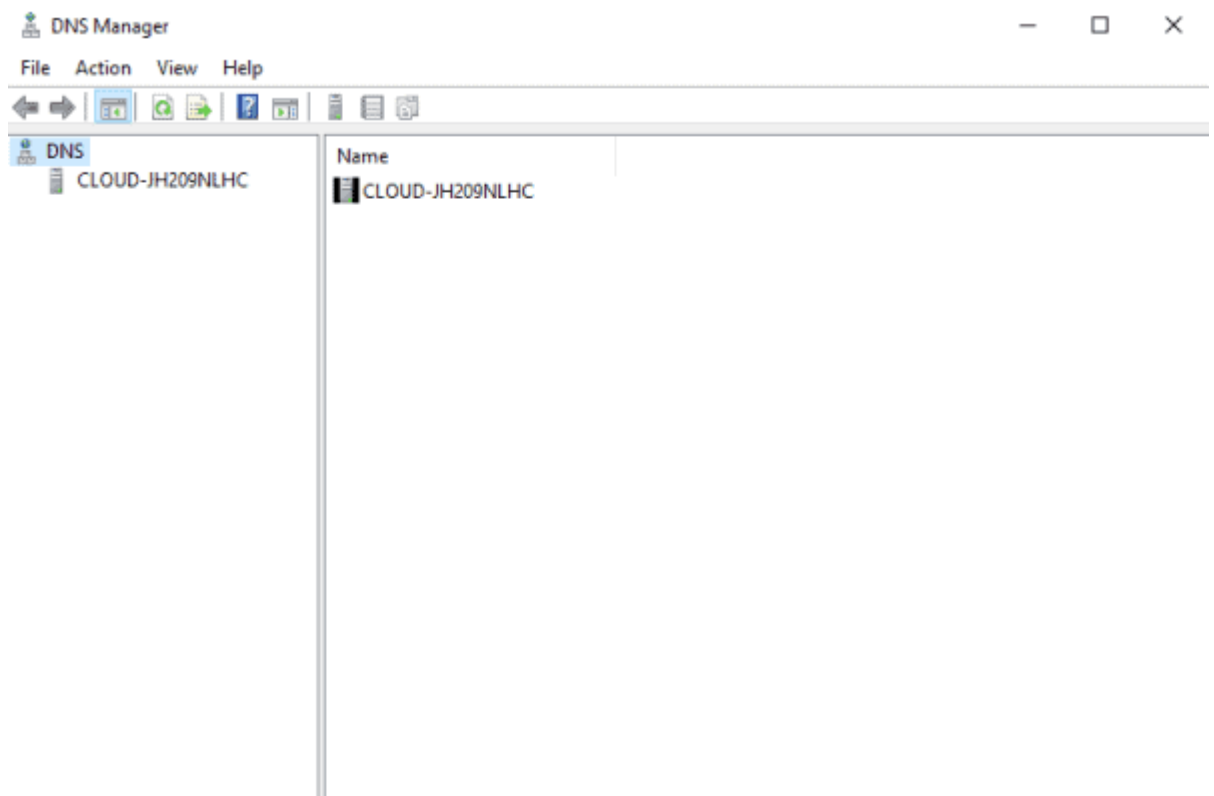
Step 1: Install DNS Role

1. Open **Server Manager**.
2. Click **Add Roles and Features**.
3. Select **Role-based or feature-based installation**.
4. Choose your server.
5. Select **DNS Server** and proceed with installation.

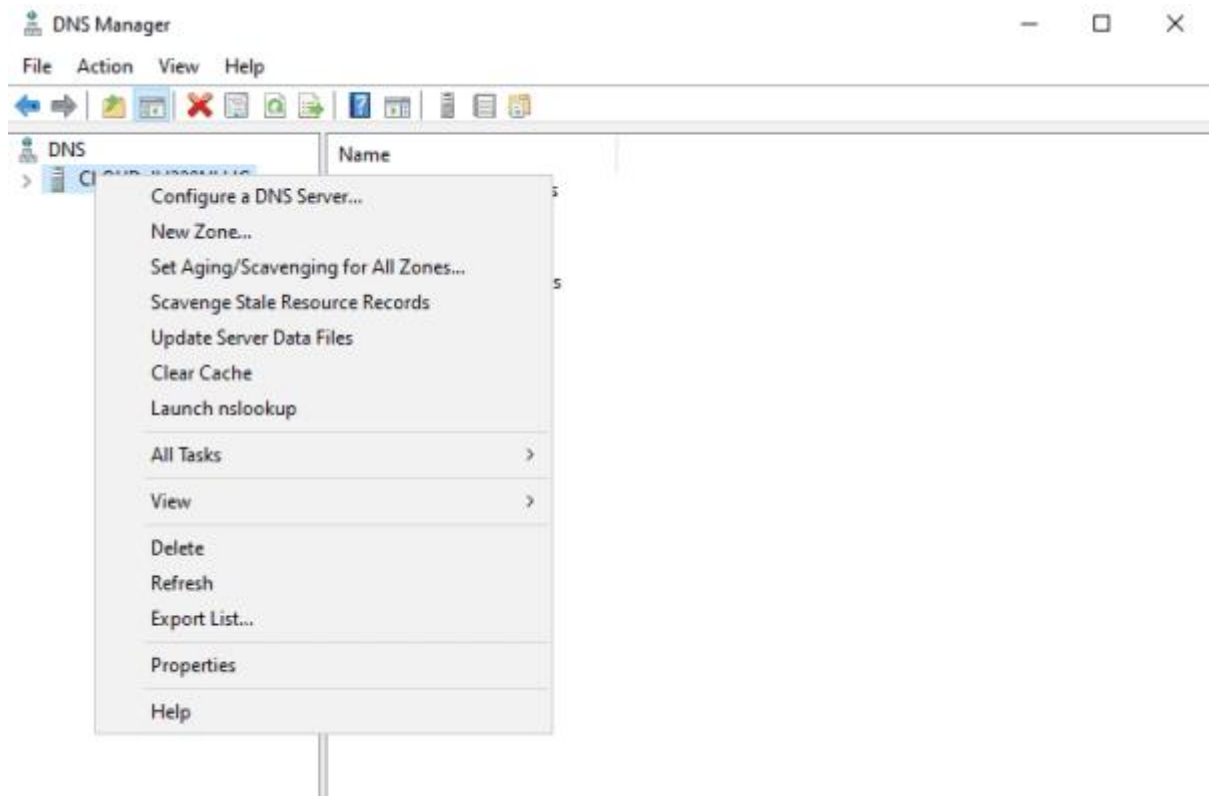


Step 2: Configure DNS Server

- **Step 1:** On the server manager, navigate to **Tools > DNS** to access the DNS manager, as shown below:



-
- **Step 2:** Right click on the server name and select **Properties**.



-
- **Step 3:** Select the **New Zone** option.
- **Step 4:** Press **Next**.



-
- **Step 5:** Choose the Primary zone and press **Next**.

New Zone Wizard



Zone Type

The DNS server supports various types of zones and storage.



Select the type of zone you want to create:

- ☒ Primary zone
Creates a copy of a zone that can be updated directly on this server.
- ☐ Secondary zone
Creates a copy of a zone that exists on another server. This option helps balance the processing load of primary servers and provides fault tolerance.
- ☐ Stub zone
Creates a copy of a zone containing only Name Server (NS), Start of Authority (SOA), and possibly glue Host (A) records. A server containing a stub zone is not authoritative for that zone.
- ☐ Store the zone in Active Directory (available only if DNS server is a writeable domain controller)

< Back

Next >

Cancel

-
- **Step 6: Click **Next** after selecting the **Forward lookup zone**.**

New Zone Wizard



Forward or Reverse Lookup Zone

You can use a zone for forward or reverse lookups.



Select the type of lookup zone you want to create:

- ☒ Forward lookup zone
A forward lookup zone translates DNS names into IP addresses and provides information about available network services.
- ☐ Reverse lookup zone
A reverse lookup zone translates IP addresses into DNS names.

< Back

Next >

Cancel

-
- **Step 7: Enter the name of our zone and press **Next**.**

New Zone Wizard



Zone Name

What is the name of the new zone?



The zone name specifies the portion of the DNS namespace for which this server is authoritative. It might be your organization's domain name (for example, microsoft.com) or a portion of the domain name (for example, newzone.microsoft.com). The zone name is not the name of the DNS server.

Zone name:

exampledns.com

< Back

Next >

Cancel

-
- **Step 8:** Choose **“Create a file with the file name”** and press **Next**.

New Zone Wizard



Zone File

You can create a new zone file or use a file copied from another DNS server.



Do you want to create a new zone file or use an existing file that you have copied from another DNS server?

☒ Create a new file with this file name:

exampledns.com.dns

☐ Use this existing file:

To use this existing file, ensure that it has been copied to the folder %SystemRoot%\system32\dns on this server, and then click Next.

< Back

Next >

Cancel

-
- **Step 9:** Check the box next to **“Do not allow dynamic update”** and click **Next**.

New Zone Wizard



Dynamic Update

You can specify that this DNS zone accepts secure, nonsecure, or no dynamic updates.



Dynamic updates enable DNS client computers to register and dynamically update their resource records with a DNS server whenever changes occur.

Select the type of dynamic updates you want to allow:

- ☐ Allow only secure dynamic updates (recommended for Active Directory)
This option is available only for Active Directory-integrated zones.
- ☐ Allow both nonsecure and secure dynamic updates
Dynamic updates of resource records are accepted from any client.
 This option is a significant security vulnerability because updates can be accepted from untrusted sources.
- ☒ Do not allow dynamic updates
Dynamic updates of resource records are not accepted by this zone. You must update these records manually.

< Back

Next >

Cancel

-
- **Step 10:** Press the **Finish** button.

New Zone Wizard



Completing the New Zone Wizard

You have successfully completed the New Zone Wizard. You specified the following settings:

Name: exampledns.com
Type: Standard Primary
Lookup type: Forward
File name: exampledns.com.dns

Note: You should now add records to the zone or ensure that records are updated dynamically. You can then verify name resolution using nslookup.

To close this wizard and create the new zone, click Finish.

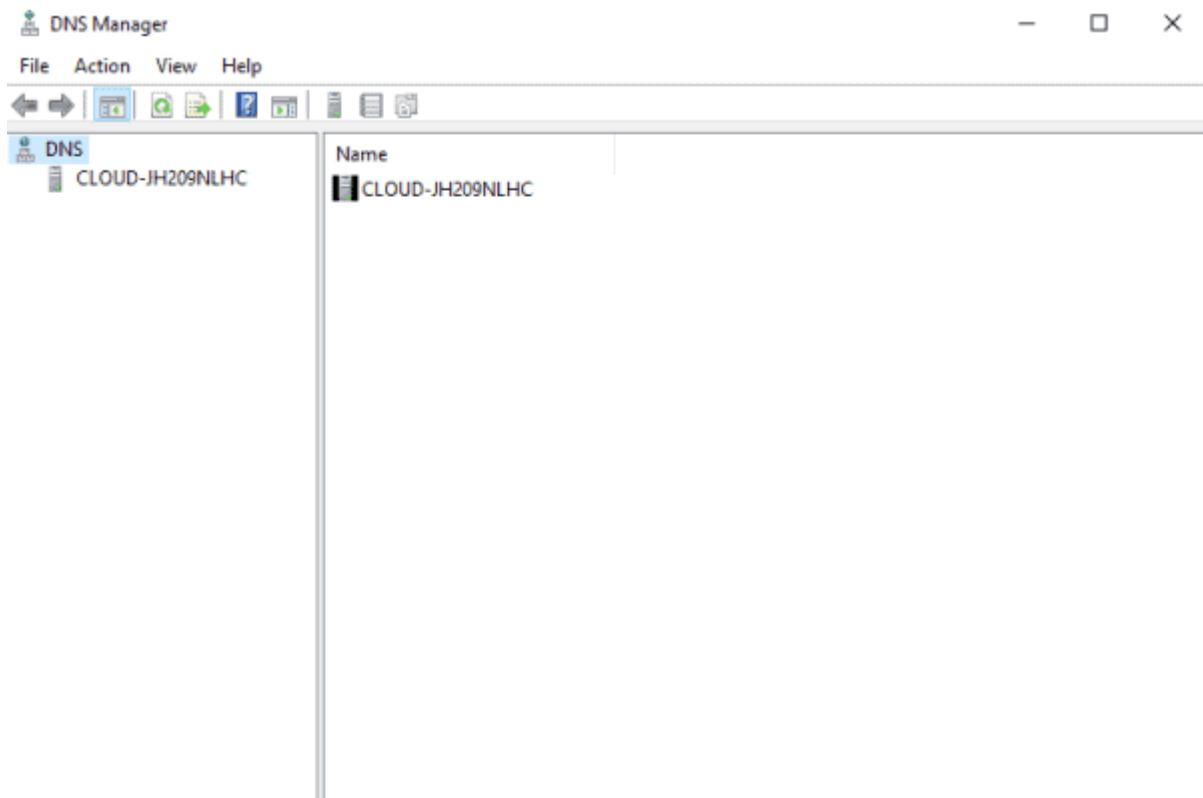
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Finish

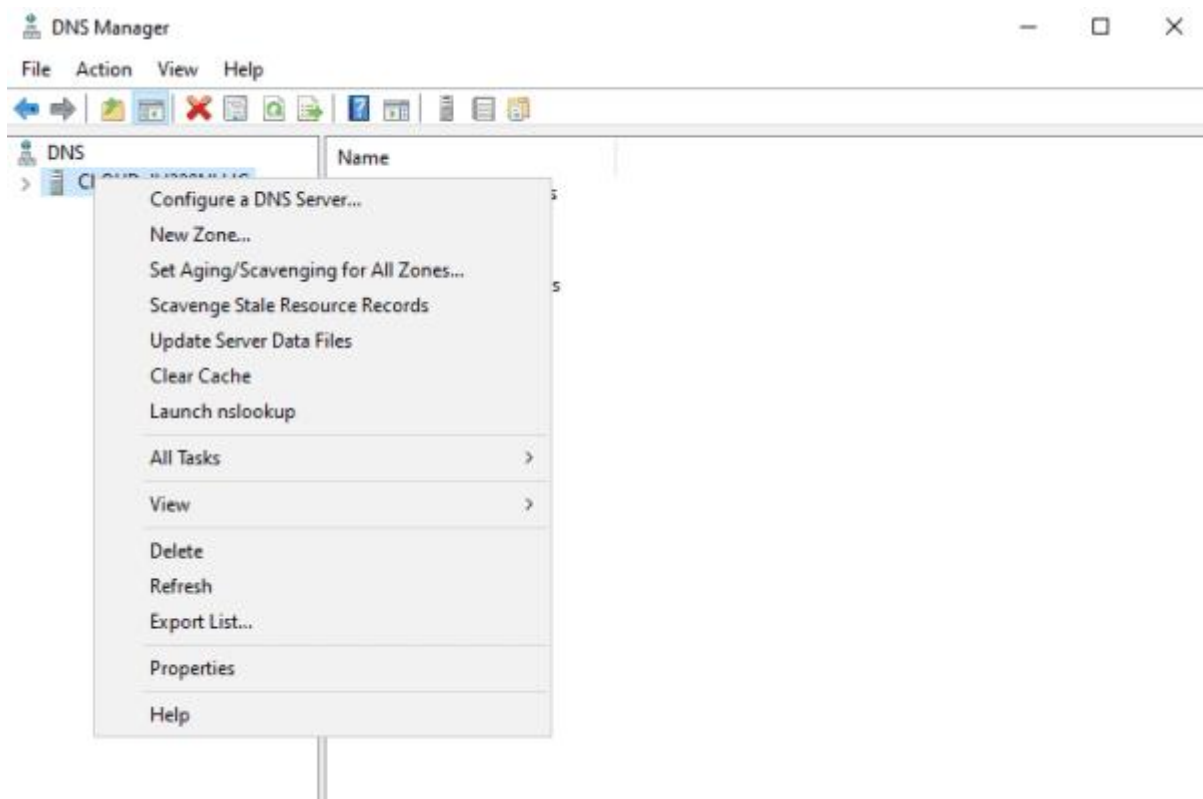
Cancel

Configure the reverse lookup also

Step 1: On the server manager, navigate to **Tools > DNS** to access the DNS manager:

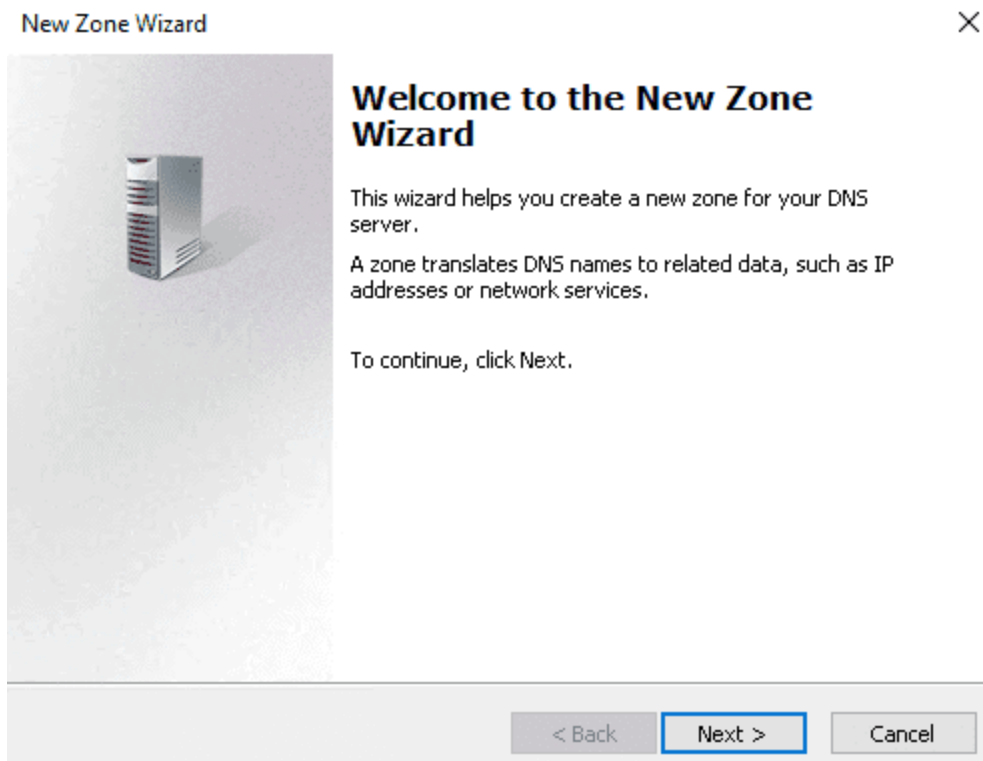


Step 2: Right click on the server name and select **Properties**.

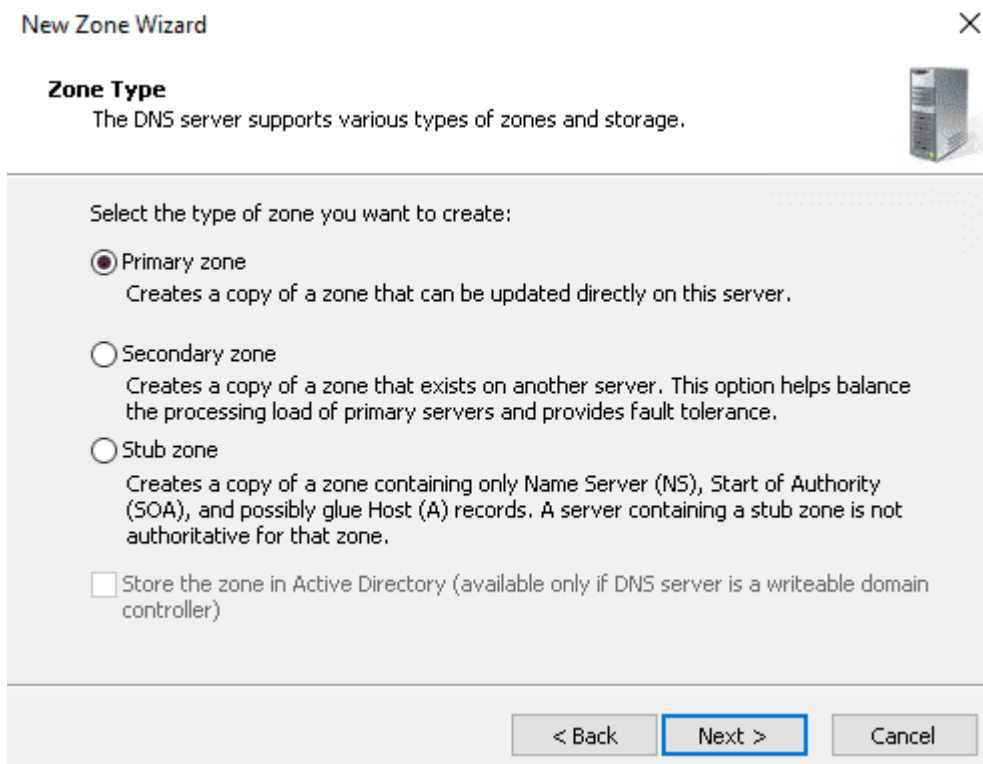


Step 3: Here, select the **New Zone** option.

Step 4: Press **Next**.



Step 5: Now, choose the Primary zone and press **Next.**



Step 6: From this step forward, the setup is different from our previous section. Next, click **Next after selecting the **Reverse lookup zone**.**

New Zone Wizard



Forward or Reverse Lookup Zone

You can use a zone for forward or reverse lookups.



Select the type of lookup zone you want to create:

☐ Forward lookup zone

A forward lookup zone translates DNS names into IP addresses and provides information about available network services.

☒ Reverse lookup zone

A reverse lookup zone translates IP addresses into DNS names.

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Next >

Cancel

Step 7: After, select **IPv4 Reverse Lookup Zone** and click on **Next**.

New Zone Wizard



Reverse Lookup Zone Name

A reverse lookup zone translates IP addresses into DNS names.



Choose whether you want to create a reverse lookup zone for IPv4 addresses or IPv6 addresses.

☒ IPv4 Reverse Lookup Zone

☐ IPv6 Reverse Lookup Zone

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Next >

Cancel

Step 8: Define your network ID and click **Next**.

New Zone Wizard



Reverse Lookup Zone Name

A reverse lookup zone translates IP addresses into DNS names.



To identify the reverse lookup zone, type the network ID or the name of the zone.

☒ Network ID:

The network ID is the portion of the IP addresses that belongs to this zone. Enter the network ID in its normal (not reversed) order.

If you use a zero in the network ID, it will appear in the zone name. For example, network ID 10 would create zone 10.in-addr.arpa, and network ID 10.0 would create zone 0.10.in-addr.arpa.

☐ Reverse lookup zone name:

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Next >

Cancel

Step 9: Further, choose “**Create a file with the file name**” and press **Next**.

New Zone Wizard



Zone File

You can create a new zone file or use a file copied from another DNS server.



Do you want to create a new zone file or use an existing file that you have copied from another DNS server?

☒ Create a new file with this file name:

☐ Use this existing file:

To use this existing file, ensure that it has been copied to the folder %SystemRoot%\system32\dns on this server, and then click Next.

< Back

Next >

Cancel

Step 9: Please check the box next to “**Do not allow dynamic update**” and click **Next**.

New Zone Wizard



Dynamic Update

You can specify that this DNS zone accepts secure, nonsecure, or no dynamic updates.



Dynamic updates enable DNS client computers to register and dynamically update their resource records with a DNS server whenever changes occur.

Select the type of dynamic updates you want to allow:

- ☐ Allow only secure dynamic updates (recommended for Active Directory)
This option is available only for Active Directory-integrated zones.
- ☐ Allow both nonsecure and secure dynamic updates
Dynamic updates of resource records are accepted from any client.
 This option is a significant security vulnerability because updates can be accepted from untrusted sources.
- ☒ Do not allow dynamic updates
Dynamic updates of resource records are not accepted by this zone. You must update these records manually.

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Next >

Cancel

Step 10: In sum, press the **Finish** button.

New Zone Wizard



Completing the New Zone Wizard

You have successfully completed the New Zone Wizard. You specified the following settings:

Name:	85.117.208.in-addr.arpa
Type:	Standard Primary
Lookup type:	Reverse
File name:	85.117.208.in-

Note: You should now add records to the zone or ensure that records are updated dynamically. You can then verify name resolution using nslookup.

To close this wizard and create the new zone, click Finish.

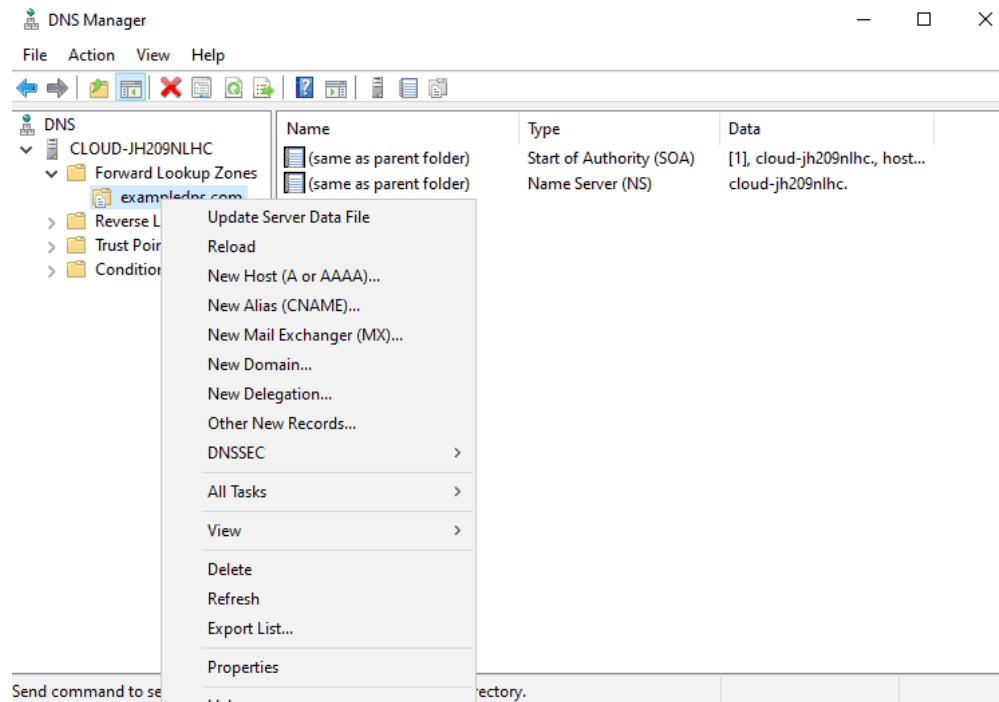
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Finish

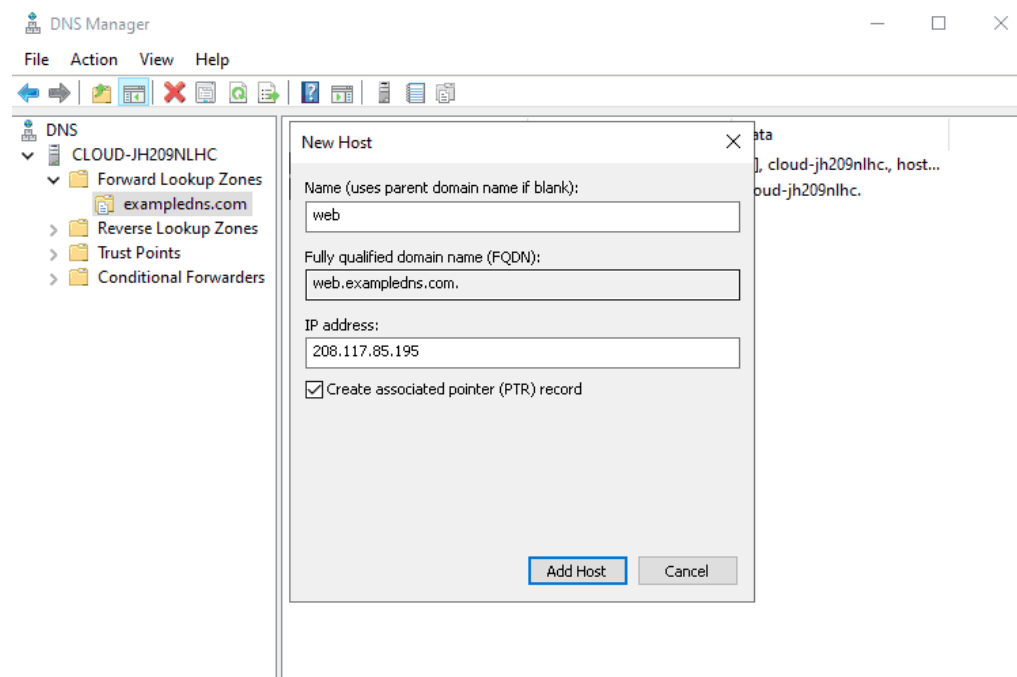
Cancel

Now we need to add host AAA

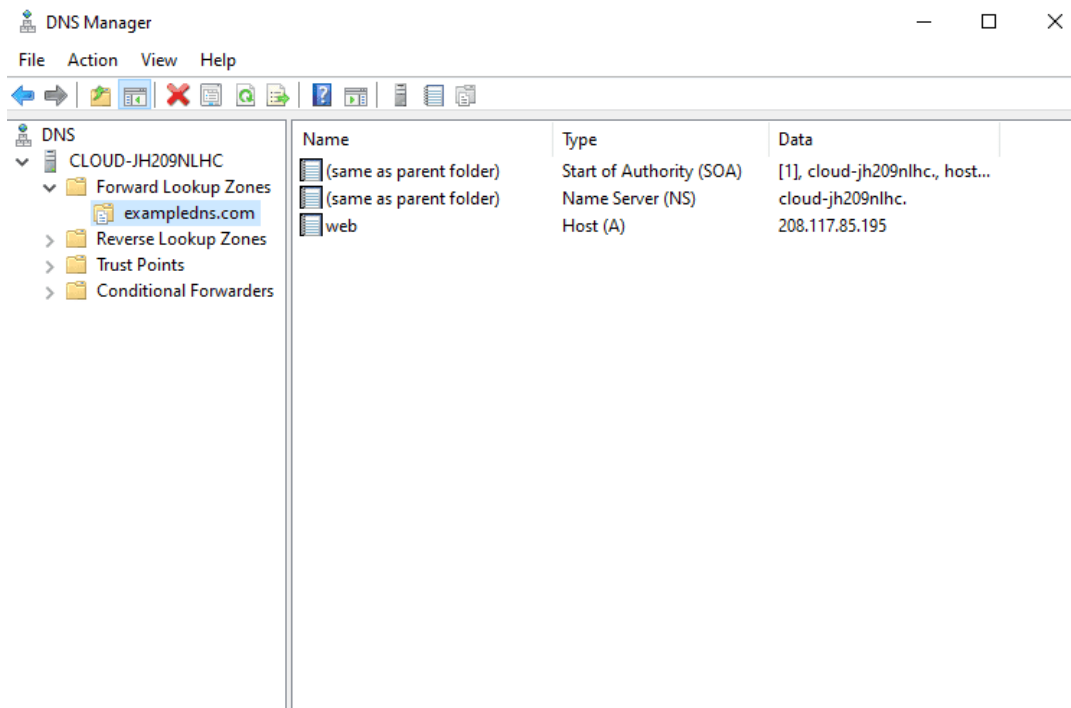
Step 1 : On the DNS manager, right click on the **forward lookup** zone.



Step 2: Click on the **New Host (A or AAAA)**.



Step 3: Provide the host's parent domain name and IP address, select “**Create associated pointer record**” and then click on the **Add Host** button.



When creating a zone in Windows Server DNS, the Start of Authority (**SOA**) and Name Server (**NS**) records are automatically created for several reasons:

Step 3: Test DNS Resolution

Now, we need to test whether the DNS name resolution works. First, open our command line interface and run the **nslookup** command:

```
nslookup
```

We should see the name of the Primary DNS server that our machine is currently connected to:

```
Default Server: web.exampledns.com
```

```
Address: 208.117.85.195
```

In fact, now please type the Fully Qualified Domain Name (FQDN) of our added host:

```
sample.exampledns.com
```

We should see the name to IP address resolution in the following output. A valid output means that our forward lookup is working:

```
Server: web.exampledns.com
```

```
Address: 208.117.85.195
```

Name: sample.exampledns.com

Address: 208.117.85.1

Subsequently, next try typing, the IP address of our added host:

208.117.85.1

And we should see the IP address to name resolution in the following output which also means our reverse lookup is functioning:

Server: web.exampledns.com

Address: 208.117.85.195

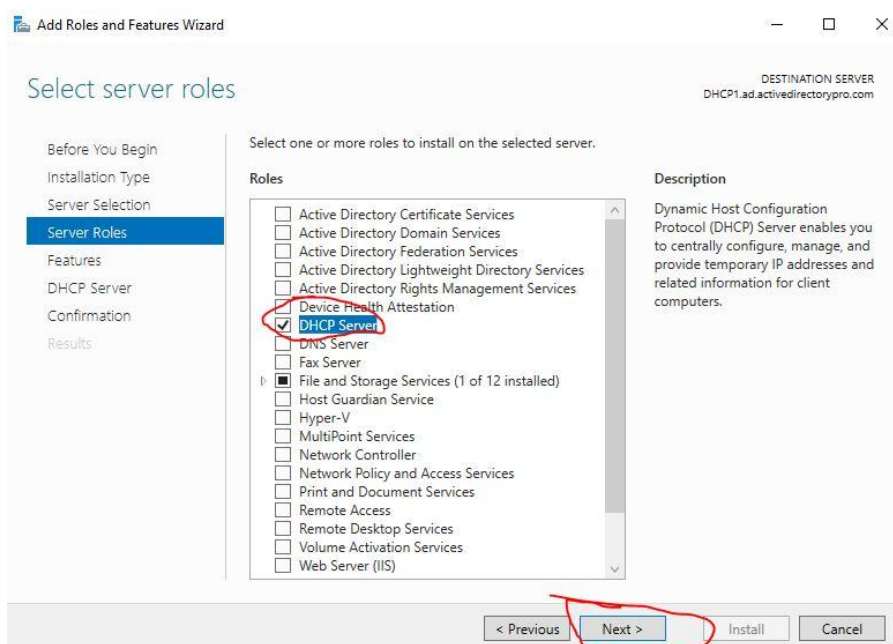
Name: sample.exampledns.com

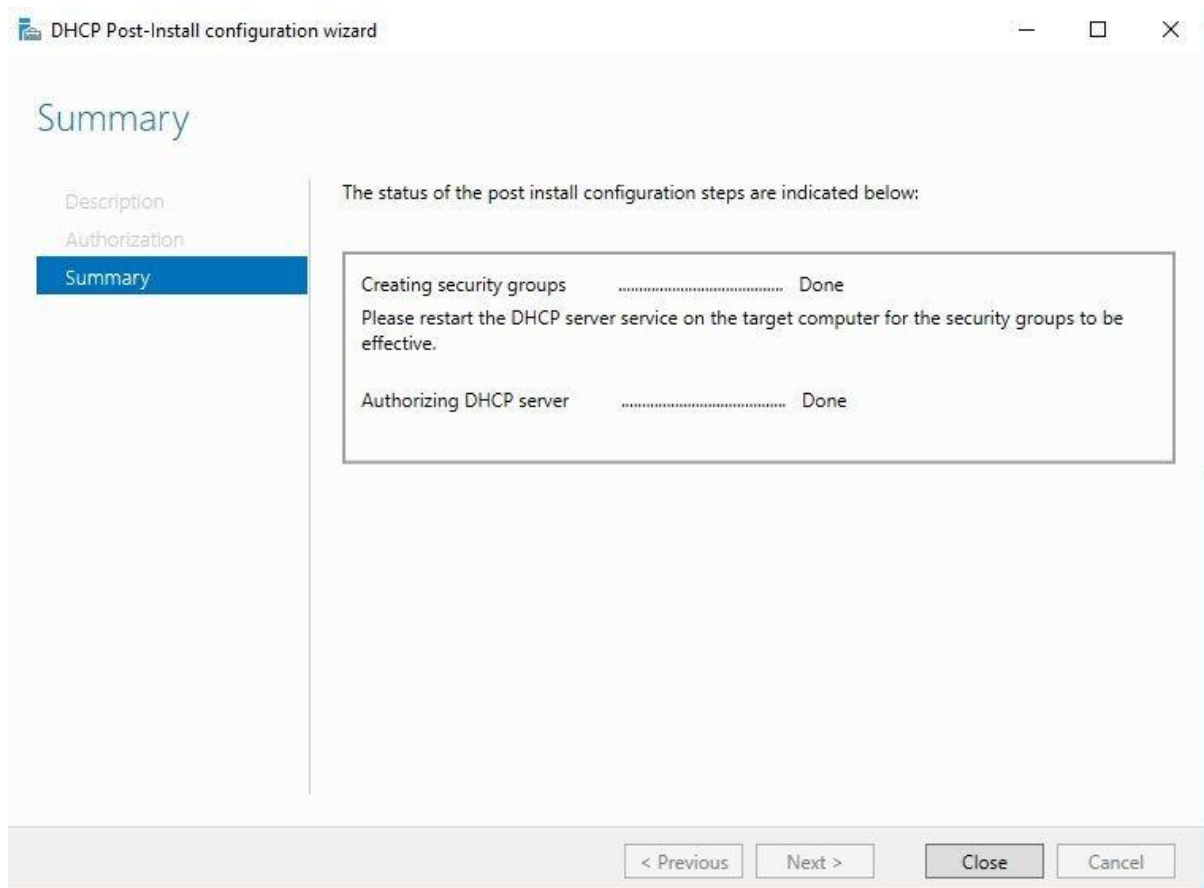
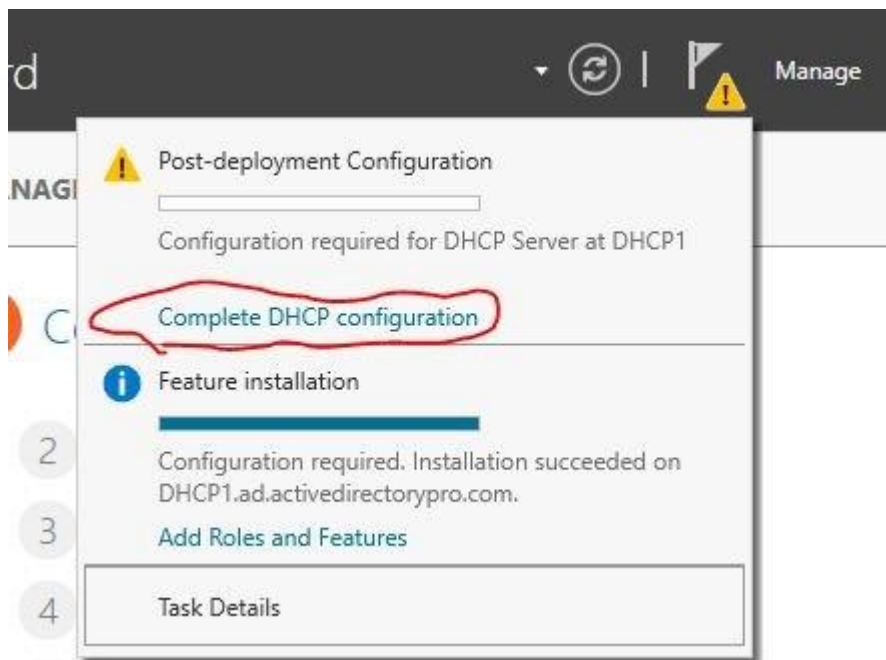
Address: 208.117.85.1

Steps to Install and Configure DHCP

Step 1: Install DHCP Role

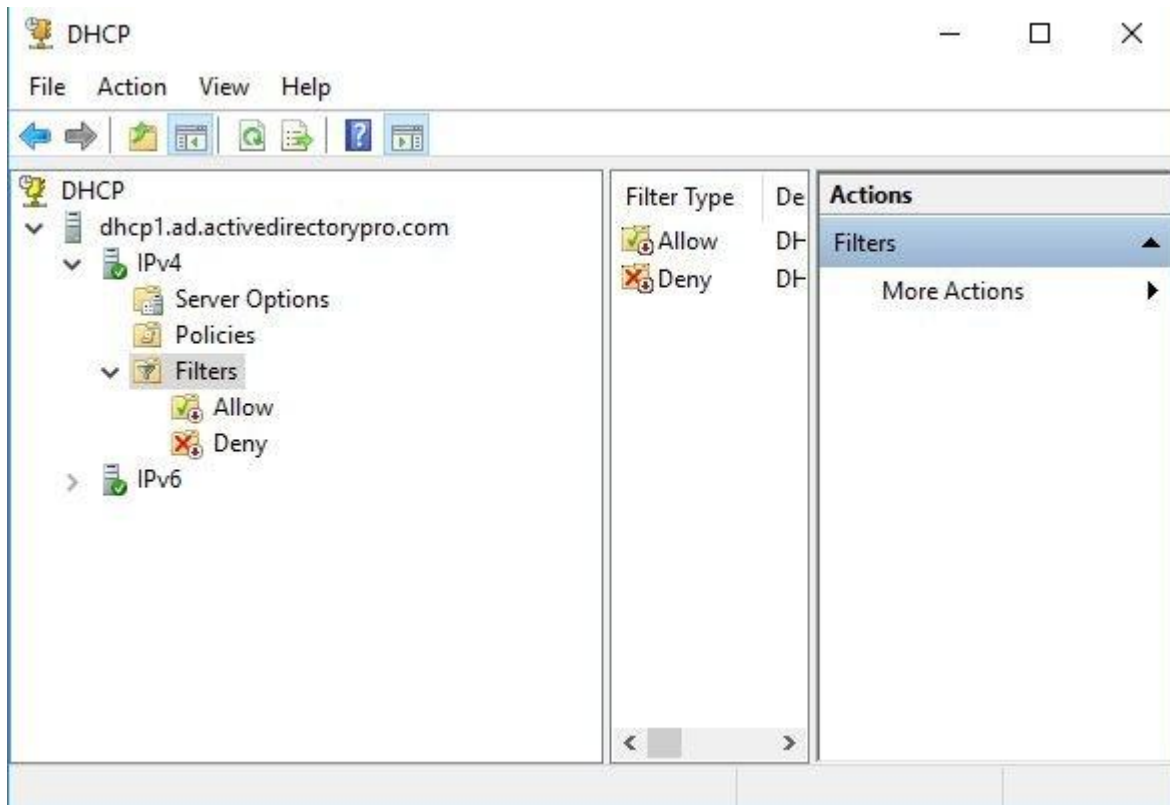
1. Open **Server Manager**.
2. Click **Add Roles and Features**.
3. Select **DHCP Server** and install it.
4. After installation, click **Complete DHCP configuration**.





Step 2: Authorize DHCP in Active Directory

1. Open **DHCP Manager** from **Tools**.
2. Right-click the server and choose **Authorize**.
3. Refresh until you see a green icon.



Step 3: Create a New DHCP Scope

1. In **DHCP Manager**, expand IPv4 → right-click and select **New Scope**.
2. Name the scope (e.g., “Office LAN”).
3. Enter IP range, e.g., 192.168.1.100 – 192.168.1.200.
4. Set subnet mask automatically.
5. Add exclusions if needed (e.g., reserve .1 for router, .10 for servers).
6. Configure lease duration (default 8 days).
7. Add DHCP options:
 - Router/Gateway: 192.168.1.1
 - DNS Server: Domain Controller’s IP (192.168.1.10)
 - Domain Name: wipro.com
8. Finish the wizard and activate the scope.

Step 4: Verify DHCP on Client

1. Join a client machine to the network.
2. Set network settings to **Obtain an IP automatically**.

3. Run `ipconfig /renew` in Command Prompt.
4. Check with `ipconfig /all` that the client received IP, gateway, and DNS from DHCP.